

Illinois Institute of Technology
Mechanical, Materials, and Aerospace Engineering
MMAE 412, Spacecraft Design, Spring 2026

Class Days/Time: MW 1:50pm-3:05pm Central Time, 1/12/26 – 5/2/26

Classroom: Stuart Building 112J

[REDACTED]

Office Location: PH-206, cubicle C-5

[REDACTED]

[REDACTED]

[REDACTED]

TA email:

[REDACTED]

Prerequisite: MMAE 411 Spacecraft Dynamics

Course Description

Spacecraft systems design including real world mission analysis and orbit design, system engineering, launch vehicle requirements, attitude determination and control, propulsion, structural design, power systems thermal management, and telecommunications.

Semester-long project is focused on the integration of multiple systems into a coherent spacecraft system to achieve specific mission requirements.

Course Outcomes

Upon the course completion, students will be able to:

- Develop an ability to integrate the knowledge and methods from previous courses on the fundamental engineering science pertaining to the design of spacecraft
- Understand the system-level considerations of the spacecraft design process, consisting of: (a) definition of mission objectives, requirements, and constraints; (b) orbit design; (c) launch vehicles and propulsion; (d) spacecraft sizing, mass, and power budgeting
- Design and/or analyze spacecraft payloads and subsystems including: (a) electrical power subsystem; (b) communications and data handling; (c) attitude determination and control; (d) thermal control
- Recognize the importance of well-functioning teams toward mission success

- Emphasize the importance of written and graphical communication for varied audiences

Course Website

Login to <https://portal.iit.edu> and follow the link for Canvas, IIT's online learning system. Then select the course MMAE 412. All course materials, links, and assignments will be posted and collected on Canvas.

Required Materials

Textbooks

- NASA *Systems Engineering Handbook* (SEH), (posted on Canvas)
- *Space Mission Analysis and Design* (SMAD), Wertz and Larson (relevant chapters posted on Canvas)
- *STK Level 1 and Level 2 Training Manual*, Ansys (STK) (posted on Canvas)

Additional Optional Readings

- Fortescue, Swinerd, and Stark, *Spacecraft Systems Engineering*, Wiley, 2011, Print ISBN:9780470750124 |Online ISBN:9781119971009 |DOI:10.1002/9781119971009
- Griffin, M. D., and J. R. French, *Space Vehicle Design, 2nd Ed.*, AIAA (American Institute of Aeronautics & Astronautics), 2004. ISBN-13: 978-1563475399. ISBN-10: 1563475391
- Pisacane, V. L., *The Space Environment and its Effects on Space Systems*, 2nd Ed., AIAA (American Institute of Aeronautics & Astronautics), 2016. ISBN-13: 978-1-62410-353-7

Other equipment / material requirements

Class will meet in a computer cluster during the term. Outside of class each student must have a device for accessing and working on course and team materials. Use of mission orbit analysis software will also be required (instructions for access will be provided).

Group meetings outside of classtime should be conducted in accordance with university policy (in-person if permitted, otherwise remotely). Use of university-provided resources (e.g., Microsoft OneDrive) is recommended. Students are permitted, but not required, to use third party software for meetings separate from classtime (e.g., skype, zoom,

Classroom Protocol

Mondays will be in-person lectures plus team work as time permits, and Wednesdays will be team labs. Students will be assigned to teams of about four people.

Arrive to class on time, because class updates will be announced at the beginning. Cell

Laptops are permitted for course-related use only. I expect everyone to treat each other with empathy and civility, upholding the student code of conduct, both in person and online.

Assignments and Grading Policy

Grading is based on the following:

Teaming homework (1 and 4)	10%
Project Assignments (2, 3, 5)	10%
Individual Quizzes	10%
Midterm	30%
Midterm Presentation	10%
Midterm Report	15%
Midterm self- & peer contribution reporting	5%
Final	35%
Final Presentation	15%
Final Report	15%
Final self- & peer contribution reporting	5%
Attendance	5%

Each quiz or project assignment (PA) will be **due online by 11:59pm Central Time on the Friday due date**, submitted via Canvas.

The overall course grade is computed on a Likert 4-point scale, similar to the rubrics that will be used to assess project assignments: Below expectations (25%), Near expectations (50%), Meets expectations (75%), Exceeds expectations (100%). Course scores will be rounded to the nearest of these scores and assigned a letter grade accordingly.

Exceeds expectations = A

Meets expectations = B

Near expectations = C

Below expectations = E

Academic Integrity Policy

Departmental policies on academic honesty are consistent with those of the University. The Department policies are provided at https://www.iit.edu/sites/default/files/2021-01/honesty_policy_-_mmae.pdf

AI Policy: AI may be used as a study tool for quizzes, but you must 1) declare on which problems it was used in your submission, and 2) submit a log of your interaction. The AI

policy for project assignments, midterm, and final will be specified at the time of the project.

Collaboration policy

The collaboration policy for this class is as follows:

Collaboration on quiz is permitted, but you must submit your own unique solutions. Any use of homework, reports, or solutions from prior years is prohibited. *If AI has ingested these, submit your logs so you do not violate this policy!* Collaboration on the project (project assignments, midterm presentation and report, and final presentation and report) is required within a team. You may choose to interact with other teams or not, at your own discretion, keeping in mind that they may be in competition with you.

Accommodations, Resources, and COVID-19 Protocols

- Reasonable accommodations will be made for students with documented disabilities. The procedure for obtaining accommodations is [here](https://www.iit.edu/cdr/services/reasonable-accommodations): <https://www.iit.edu/cdr/services/reasonable-accommodations>
- Additional learning resources available at IIT are [here](https://www.iit.edu/sites/default/files/2021-08/additional-resources-COVID-19-classroom-protocols.pdf): <https://www.iit.edu/sites/default/files/2021-08/additional-resources-COVID-19-classroom-protocols.pdf>